DIVP-TA

Q1 Consider the hand X-ray images shown below. The image on the right was obtained by low pass filtering the image on the left with a Gaussian low pass filter & then highpass filtering the result with a Gaussian highpass filter. The images are of size 420×344 pixels & $D_0 = 25$ was used for both filter transfer

a) Explain why the center part of finger ring in figure on the right appears so bright & solid. Considering that dominant characteristic of filtered image consist of edges of filtered images consists of edges of fingers & wrist bones with darker areas in b/w. In other words would you not expect the highpass filter to render the constant area inside the ring as dark since a highpass filter eliminate the term & reduces low frequencies?

b) Do you think the result would have been different if the order of filtering process had been reversed?

→ 1. Understand the original image

The original image contains different intensity regions

- Bones & edges have changing intensity

- The center of ring contains relatively constant intensity region
- The background & surrounding tissues have their own intensity values

2. Apply Gaussian low-pass filter

$$f^n :- \quad H_{LP}(u,v) = e^{-\frac{D^2(u,v)}{2D_0^2}}$$

where,

$$D(u,v) = \sqrt{(u-M/2)^2 + (v-N/2)^2}$$

Since $D_0 = 25$

$$H_{LP}(u,v) = e^{\frac{-D^2(u,v)}{1250}} \quad \ldots 2(25)^2$$

The low pass filter suppress high freq components & preserve low frequency component As a result the image becomes smoother & smaller details & sharp transitions are reduced

3. Apply Gaussian high-pass filter

$$H_{HP}(u,v) = 1 - H_{LP}(u,v)$$

$$H_{HP}(u,v) = 1 - e^{\frac{D^2(u,v)}{1250}}$$

Step1 Examine the ring in original image
The ring is a small highly contrasted object against the surrounding finger. Its brightness produces strong intensity transitions.

Step2 Apply Gaussian low pass filter
The low pass filter smooths the rings boundaries & spreads some of its intensity into neighbouring pixels. The rings sharp edges become blurred.

Step3 Apply the Gaussian high pass filter
The high pass filter suppresses the low freq components of this blurred image & retains some of its more rapid intensity variations

Step4 why is the center not necessarily black?
1. The center of ring looks approximately constant
2. A high pass filter removes constant components
3. Therefore center must become black

final The ring appears bright and solid because it is a small high-contrast object with strong boundaries. Gaussian low pass filtering blurs & separate its intensity. while subsequent Gaussian high pass filtering retains some of the spatial-frequency components associated with the ring & its boundaries. Their combined response can make central region appear bright & filled.

(6) The originally No the result would be same assuming ideal linear filtering with same transfer f^ & no intermediate non-linear operation.

1 Low-pass followed by High-Pass

$G_1(u,v) = F(u,v)\, H_{LP}(u,v)$ → apply low pass

& $R_1(u,v) = F(u,v)\, H_{LP}(u,v)\, H_{HP}(u,v)$ → apply high pass

2 High pass followed by low pass

$G_2(u,v) = F(u,v)\, H_{HP}(u,v)$ → apply High pass first

& $R_2(u,v) = F(u,v)\, H_{HP}(u,v)\, H_{HP}(u,v)$ → apply ~~high~~ low pass

3. compare both results, multiplication is commutative

$H_{LP}(u,v)\, H_{HP}(u,v) = H_{HP}(u,v)\, H_{LP}(u,v)$

$R_1(u,v) = R_2(u,v)$

Take inverse fourier Transform $r_1(x,y) = r_2(x,y)$

Q2 Consider 3×3 spatial kernel that averages the four closest neighbours of a point (x,y) but excludes the point itself from the average

a. Find the equivalent filter transfer function $H(u,v)$ in the frequency domain

b. Show that your result a lowpass filter transform

⇒ The spatial filter kernel $h(x,y)$ of size 3×3 averages the four neighbours of pixel (x,y) excluding the pixel itself. Centering the origin $(0,0)$ at middle of the kernel the non-zero coefficients at positions (x,y) for $x,y \in (-1,0,1)$ are

$$h(x,y) = \begin{cases} 1/4 & \text{for } (x,y) \in \{(0,1)(0,-1)(1,0)(-1,0)\} \\ 0 & \text{, otherwise} \end{cases}$$

$\therefore$ 3x3 neighbors $\begin{bmatrix} 0 & 1/4 & 0 \\ 1/4 & 0 & 1/4 \\ 0 & 1/4 & 0 \end{bmatrix}$

a) find frequency domain transfer function $H(u,v)$
The 2D DFT of spatial kernel $h(x,y)$ of size $M \times N$ centered at $x,y = 0$ given by

$$H(u,v) = \sum_{x=-1}^{1} \sum_{y=-1}^{1} h(x,y) e^{-j2\pi \left(\frac{ux}{M} + \frac{vy}{N}\right)}$$

for a 3x3 spatial filter $(M=3, N=3)$

$$H(u,v) = \frac{1}{4}\left(e^{-j\frac{2\pi u}{3}} + e^{j\frac{2\pi v}{3}} + e^{-j\frac{2\pi v}{3}} + e^{j\frac{2\pi v}{3}}\right)$$

Using eulers identity $\cos(\omega) = \frac{e^{j\omega} + e^{-j\omega}}{2}$

$$H(u,v) = \frac{1}{2}\left[\cos\left(\frac{2\pi u}{3}\right) + \cos\left(\frac{2\pi v}{3}\right)\right]$$

b) Proof of bocopass filter Properly

To shoco thad $H(u,v)$ is a lowpass filter transfer function we evaluate its behaviour at key frequency Points.

1. At the DC component (origin / low frequencis, $u=0, v=0$)

$$H(0,0) = \frac{1}{2}[\cos(0) + \cos(0)]$$

$$= \frac{1}{2}(1+1)$$

$$= 1$$

It filter pass low (zero) frequency signals completely without attenuation.

2. At higher frequencies away from origin.

as frequencies u & v increases towards the edges of frequency spectrum (e.g $u = \pm 1; v = 1$)

$$H(1,1) = \frac{1}{2}\left[\cos\left(\frac{2\pi}{3}\right) + \cos\left(\frac{2\pi}{3}\right)\right] = \frac{1}{2}\left(\frac{-1}{2} - \frac{1}{2}\right) = \frac{-1}{2}$$

The magnitude $|H(u,v)| = \frac{1}{2}$ at high freq is lower than the peak value of 1 at original, attenuating high freq. components relative to low freq.

Q3. Consider the images shown. The image on the right was obtained by a) multiplying the image on the left by $(-1)^{x+y}$; (b) computing the DFT; (c) taking complex conjugate of transform d) computing inverse DFT & e) multiplying the real part of result by $(-1)^{x+y}$. Explain (mathematically) why the image on right appears as it does

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→ 1. Multiply original image by $(-1)^{x+y}$

Let original image be $f(x,y)$

$$\therefore g(x,y) = (-1)^{x+y} f(x,y)$$

2. Compute The 2D DFT

$$G(u,v) = \sum_{x=0}^{M-1} \sum_{y=0}^{N-1} g(x,y)\, e^{-j2\pi\left(\frac{ux}{M} + \frac{vy}{N}\right)}$$

where x, y = spatial coordinates
u, v = frequency coordinates
M, N = image dimensions
$j = \sqrt{-1}$

$$\Rightarrow G(u,v) = \sum_{x=0}^{M-1} \sum_{y=0}^{N-1} (-1)^{x+y} f(x,y)\, e^{-j2\pi\left(\frac{ux}{M} + \frac{vy}{N}\right)}$$

3. Take complex conjugate of $G(u,v)$

we know $G(u,v) = a + jb$
Then $G(u,v) = a - jb$

4. Compute the inverse DFT

$$g_1(x,y) = \frac{1}{MN} \sum_{u=0}^{M-1} \sum_{v=0}^{N-1} G(u,v)\, e^{j2\pi\left(\frac{ux}{M} + \frac{vy}{N}\right)}$$

Using conjugate property $g_1(x,y) = g^*(-x,-y)$
$\Rightarrow g_1(x,y) = g(-x,-y)$

Here the image is reversed along both horizontal & vertical co-ordinates.

5. Multiply real part by $(-1)^{x+y}$
final image is $h(x,y) = (-1)^{x+y} g_1(x,y)$

$\Rightarrow h(x,y) = (-1)^{x+y} g(-x,-y)$

We know,

$$g(x,y) = (-1)^{x+y} f(x,y)$$
$$g(-x,-y) = (-1)^{-x-y} f(-x,-y)$$

Put into final eqn $h(x,y) = (-1)^{x+y}(-1)^{-x-y} f(-x,-y)$

$$\therefore h(x,y) = (-1)^{x+y-x-y} f(-x,-y)$$

$$\text{i.e. } h(x,y) = (-1)^0 f(-x,-y) \quad \& \quad (-1)^0 = 1$$

$$\therefore \boxed{h(x,y) = f(-x,-y)}$$